

UNIT3: CAPACITANCE

capacitors



1- capacitor and capacitance

- concept and application
- how to compute C. $C = Q/V$
- isolated sphere capacitor
- spherical capacitor
- cylindrical capacitor / demo
- parallel plates capacitor / demo

2- dielectric

- types of capacitors used in electronics
- new equation
- concepts

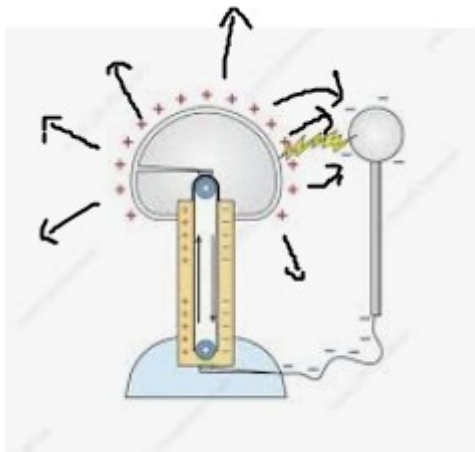
3- energy in a capacitor

- equation
- energy density

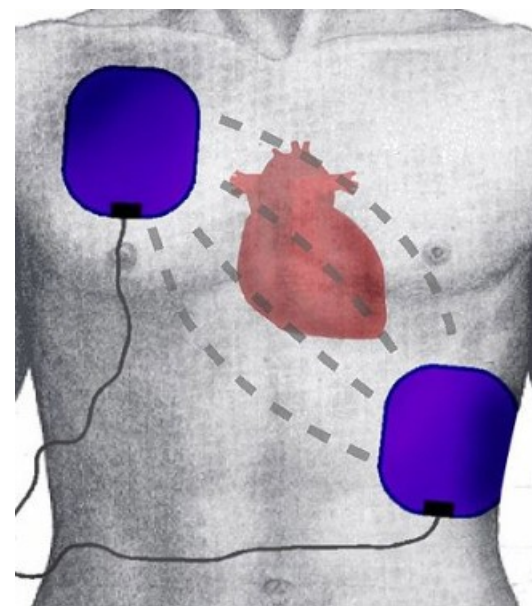
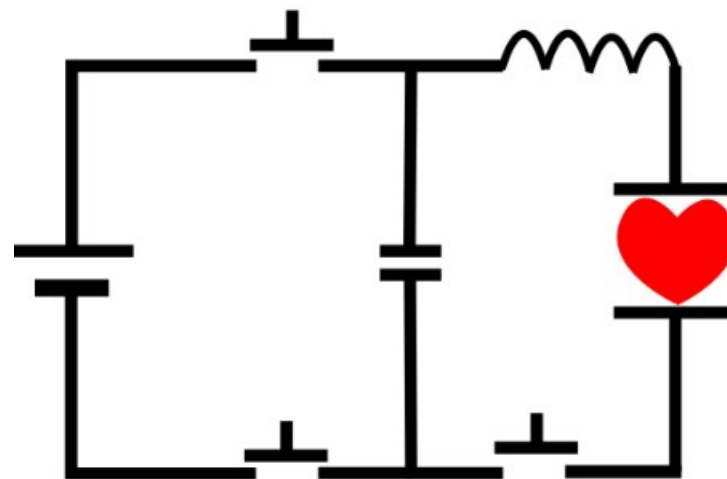
4- capacitor in parallel and in series

Capacitors can store and provide an abundant amount of energy that cannot easily be supplied by a traditional battery. Here a defibrillator.

<http://hyperphysics.phy-astr.gsu.edu/hbase/electric/capcon.html#c1>



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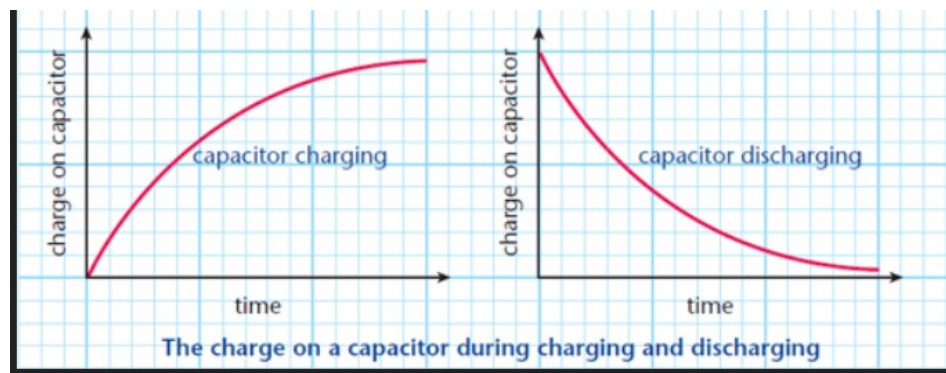


200V → 1000V for heart.

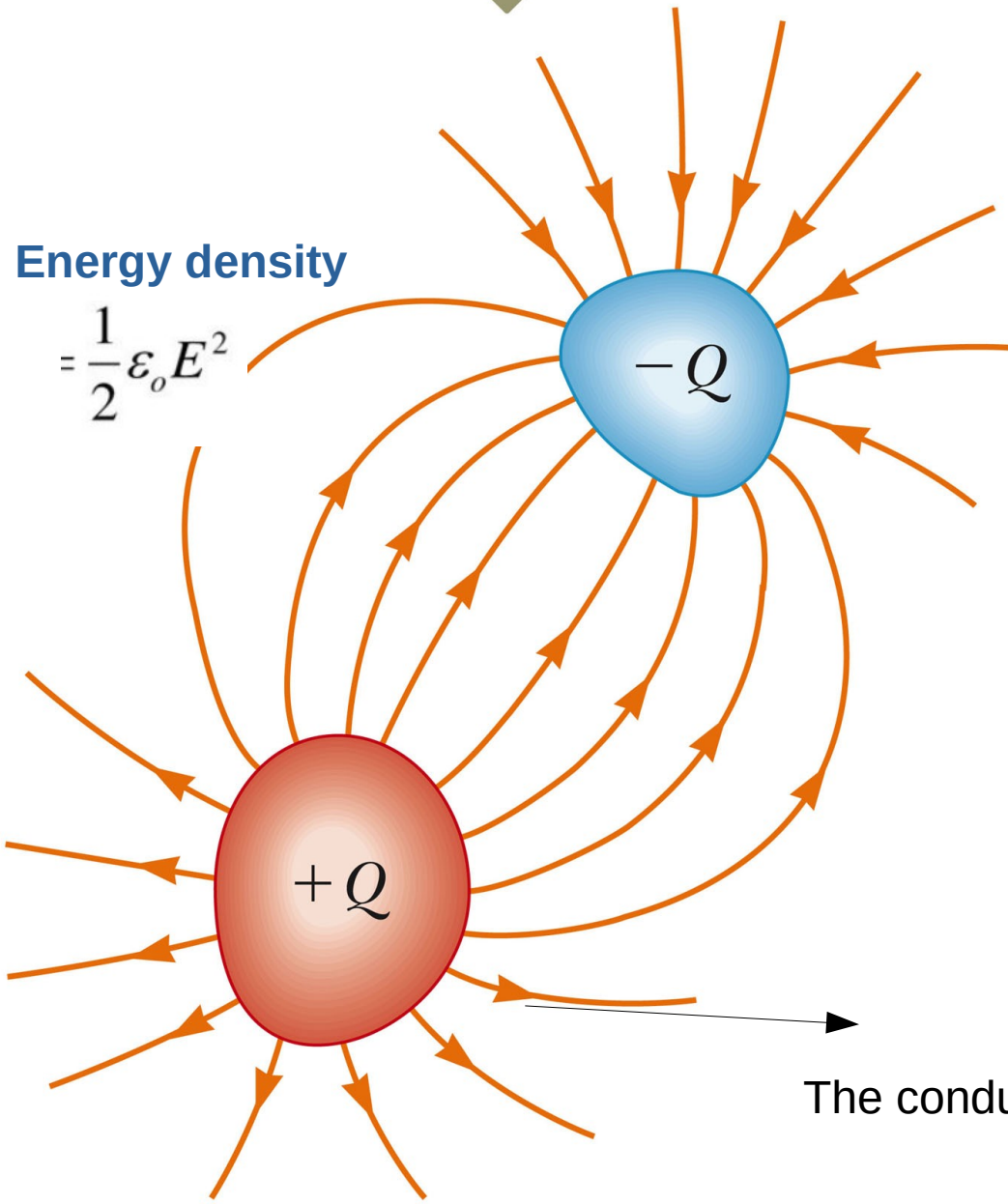
<http://micro.magnet.fsu.edu/electromag/java/capacitor/index.html>

https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html

demonstration



When the capacitor is charged, the conductors carry charges of equal magnitude and opposite sign.



Potential energy can be stored in a Stretched spring.

A book is lifted. The energy is Stored in the gravitational field of book+Earth system.

A capacitor is a device to package energy in an electrostatic field. In a storehouse.

A capacitor = 2 conductors with Opposite charges. The conductors Are called plates.

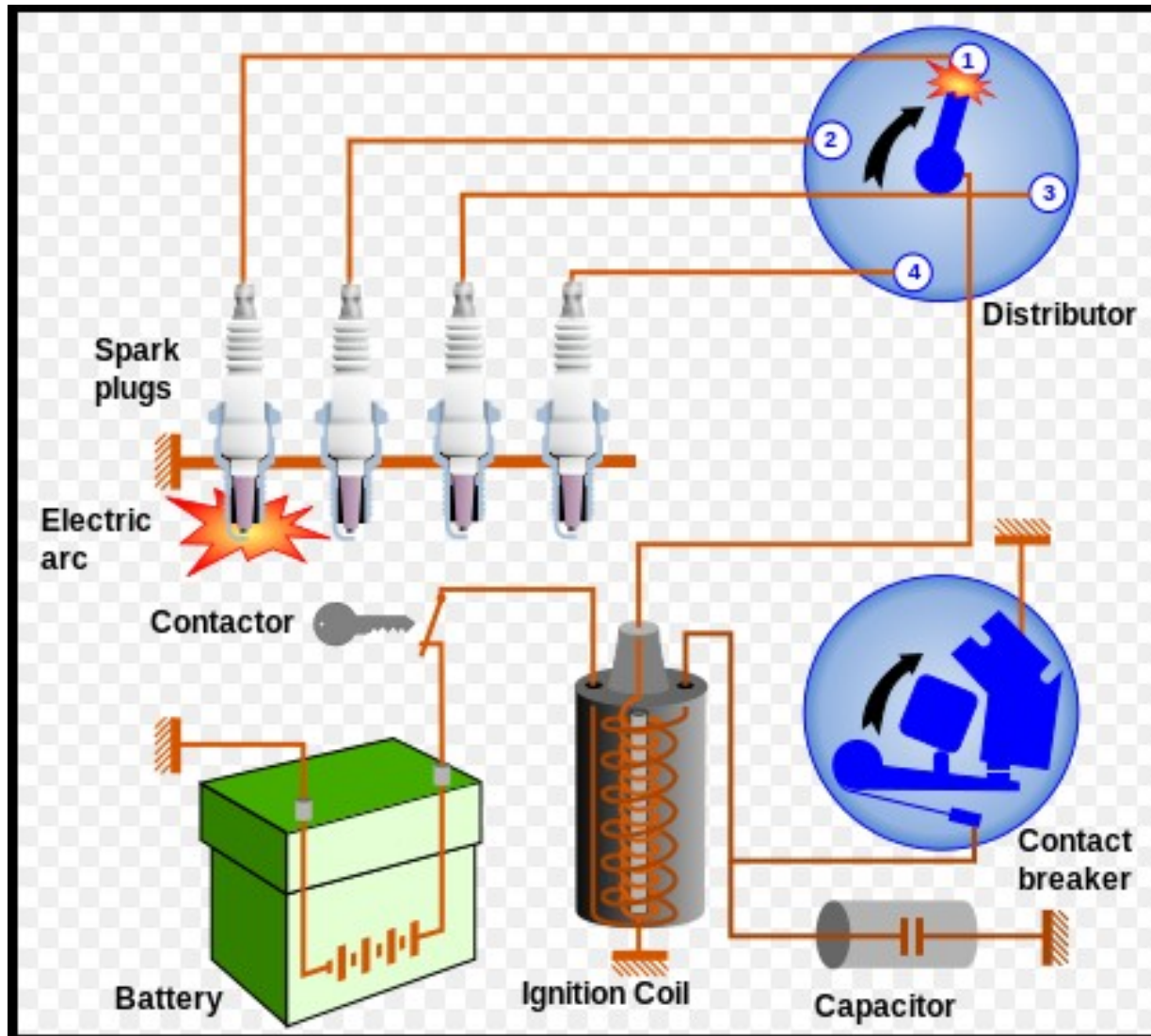
$$U = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV = \frac{1}{2} CV^2$$

This energy is stored in the electric field.

The conductor is an Equipotential



Harold Eugene Edgerton (MIT) developed high speed photography
photoflash / stroboscope
Flash of a camera : the electric energy is released as light.



Capacitor discharge ignition
 The capacitor
 Charge the inductor that
 discharge and make a
 spark.



Hindenburg disaster
 Before WWII
 Hydrogen balloon was probably charged
 And it discharged when it got close to a metallic
 Pole. Capacitors can be deadly.



RAM

Capacitors are used to store memory in a computer

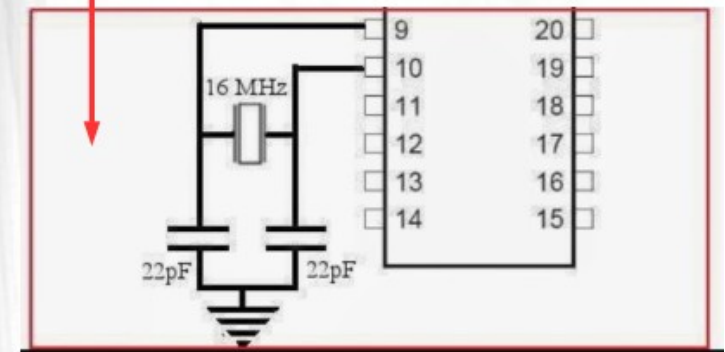


Bart in Francisco. Breaking →
KE stores in capacitors →
Electric energy back to electric batteries

Cap: 22nF for resonator
(ceramic – to reach resonance)

DC compon
of power

FIGURE 12.1



Smooth out power supply.

Remove the high frequency noise.
Low pass filters.

Caps: 100nF close to Mc to filter noise (called decoupling capacitors)
- 47 uF for power supply for 5V regulator

The capacitors are used to reduced noise.
Low pass filter. They are called decoupling
Or by pass capacitors.
Small capacitors are used with microprocessor.
Larger capacitors are used for he power supply.
(see ArduinoUno_Annotated_Voltage_Reg1.png)

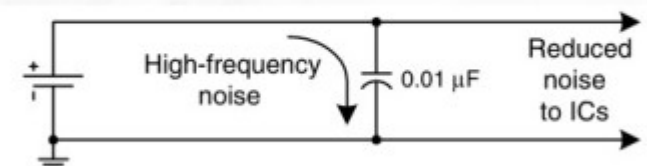
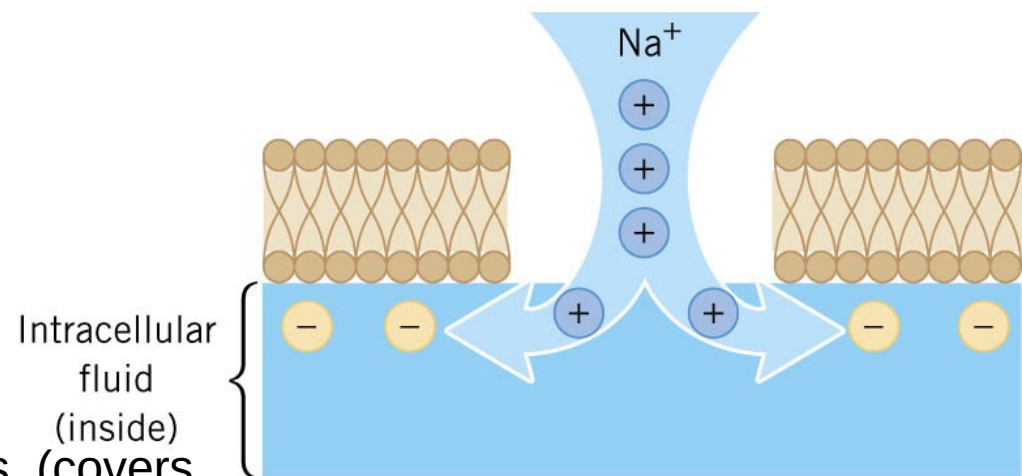
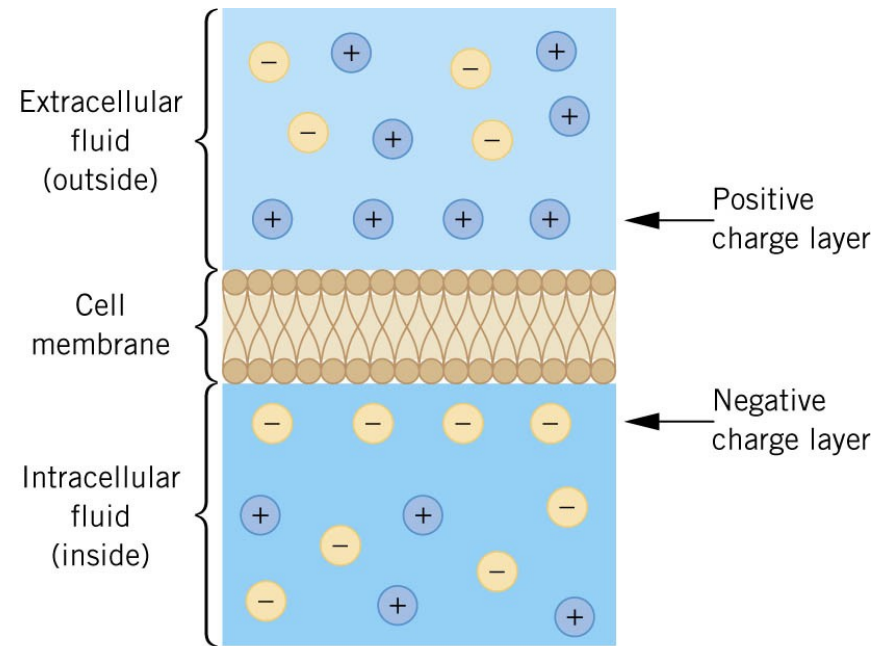
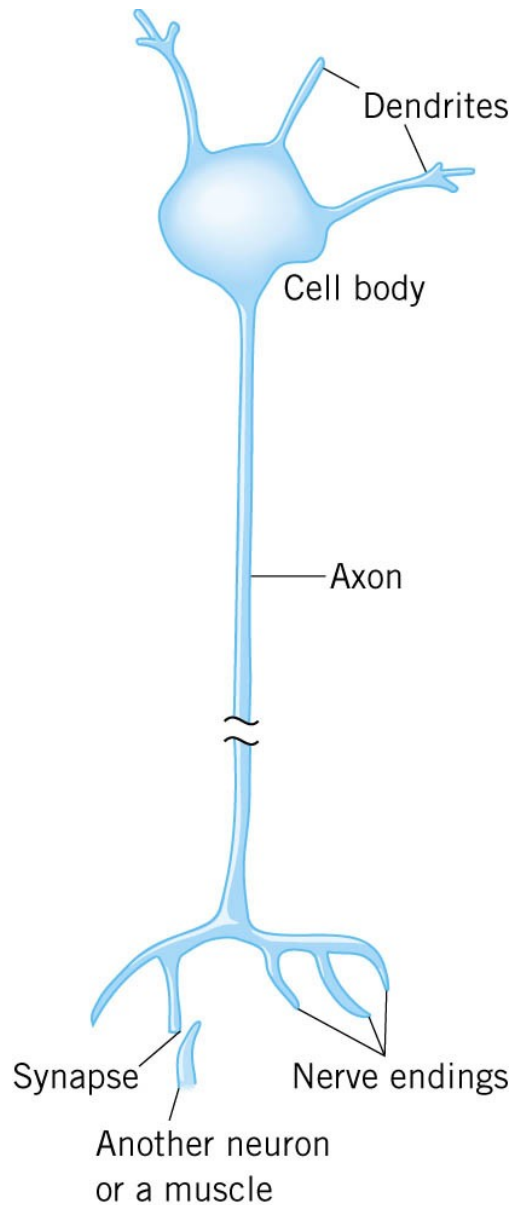


FIGURE 12.8 Noise filtering with shunt capacitance.

19.6 Biomedical Applications of Electrical Potential Differences



Great channel for medical students. (covers bio medical stuff)

<https://www.youtube.com/c/USMLEOnline>

Simulation !

THE RELATION BETWEEN CHARGE AND POTENTIAL DIFFERENCE FOR A CAPACITOR

The magnitude of the charge in each place of the capacitor is directly proportional to the magnitude of the potential difference between the plates.

$$q = CV$$

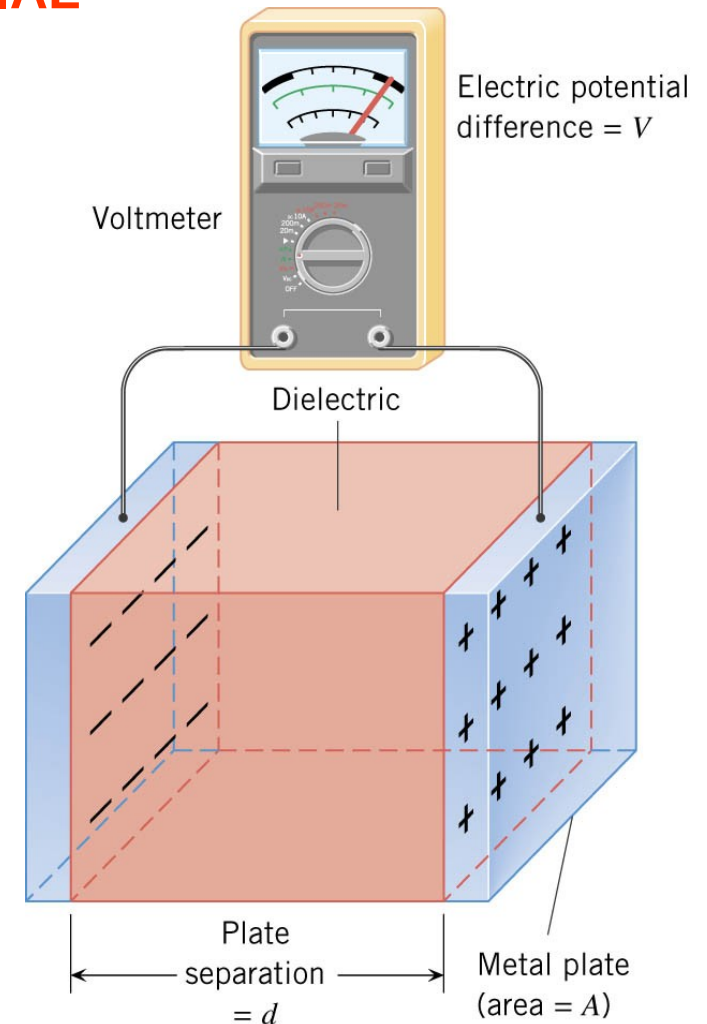
The capacitance C is the proportionality constant.

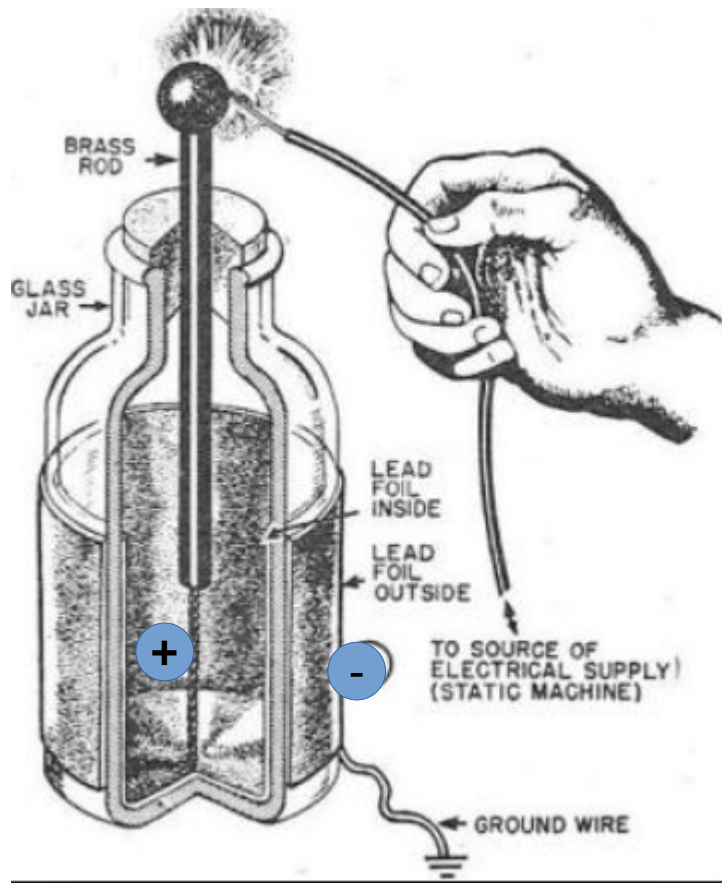
SI Unit of Capacitance: coulomb/volt = farad (F)

Think of capacity. The capacitance is the charge Per unit volt. So if you charge a 4pF with a 9V Battery then the maximum charge is 9x4pC

The charge is proportional to the voltage.

The dielectric increases the capacitance by a factor k the dielectric constant.





“Battery” of Leyden Jars

- (1747?) Daniel Gralath was the first to combine several jars in parallel into a "battery" to increase the total possible stored charge.
- He demonstrated its effects on a chain of 20 persons

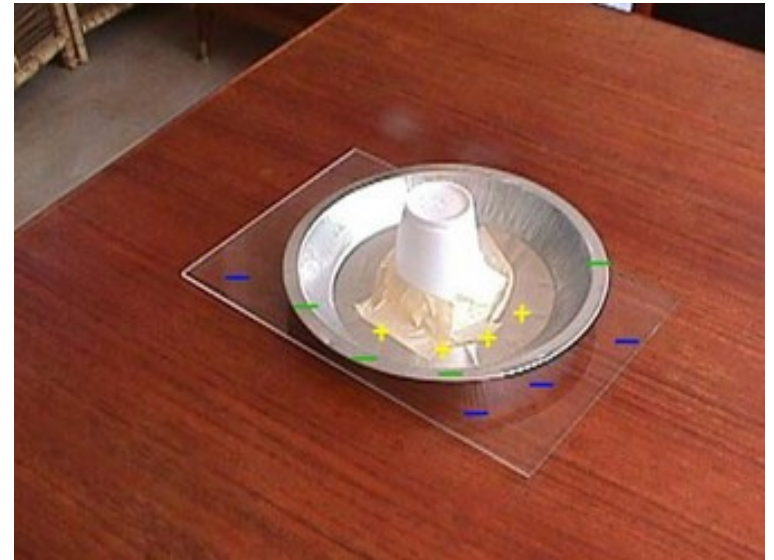


A leyden jar (1745) was one of the first capacitors. It is a simple cylindrical capacitor. (Dutch physicist Pieter Van Musschenbroek Invented the jar). Named after the University of Leyden where he worked. Daniel Gralath was the first to combine several jars in parallel into a "battery" to increase the charge storage capacity.

Film canister



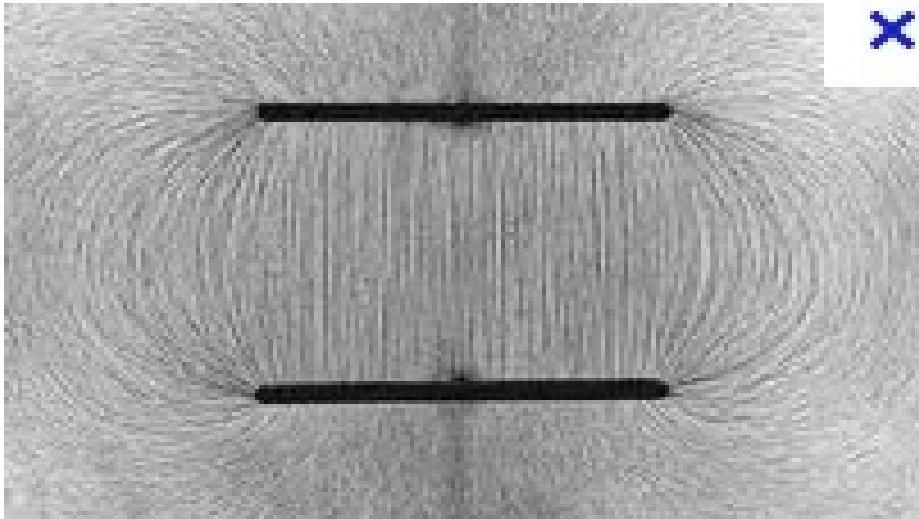
Home made capacitor (cylindrical capacitor)



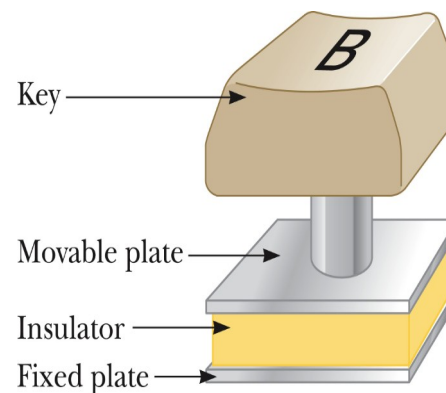
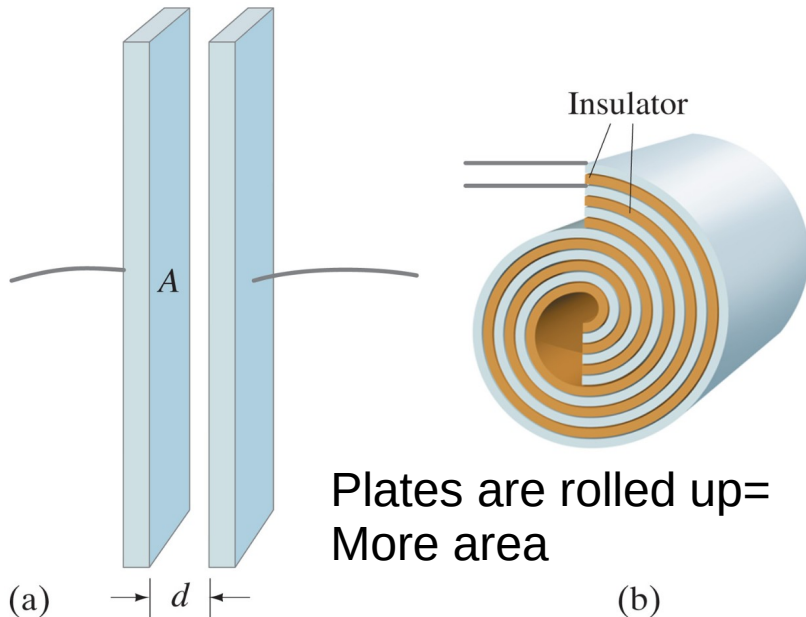
You can charge it with an Electrophorus or a Van graaf



Example: parallel plate capacitor. Electric field is homogeneous between the plates.
(we neglect fringe effect shown here).



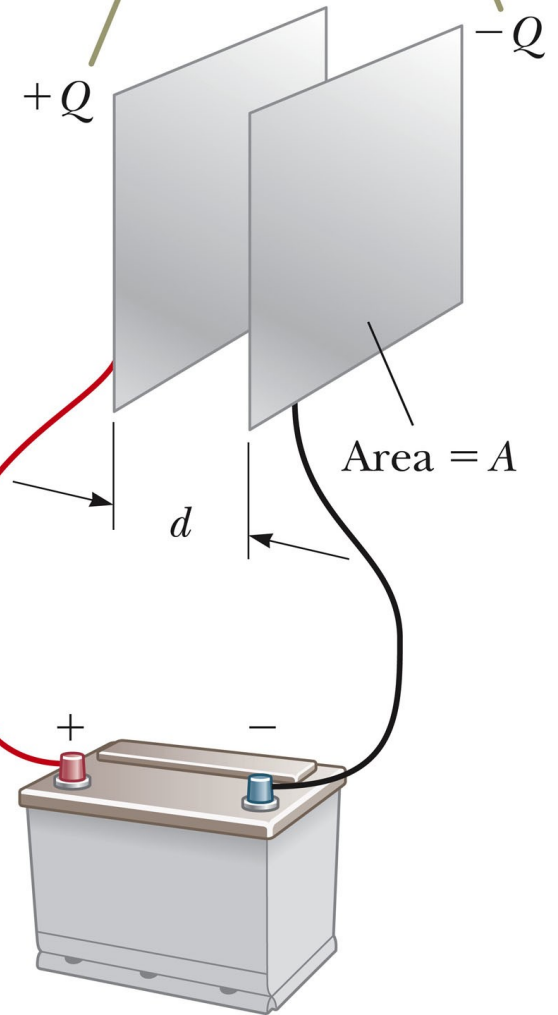
<https://phet.colorado.edu/en/simulation/capacitor-lab>



Application: keyboard.

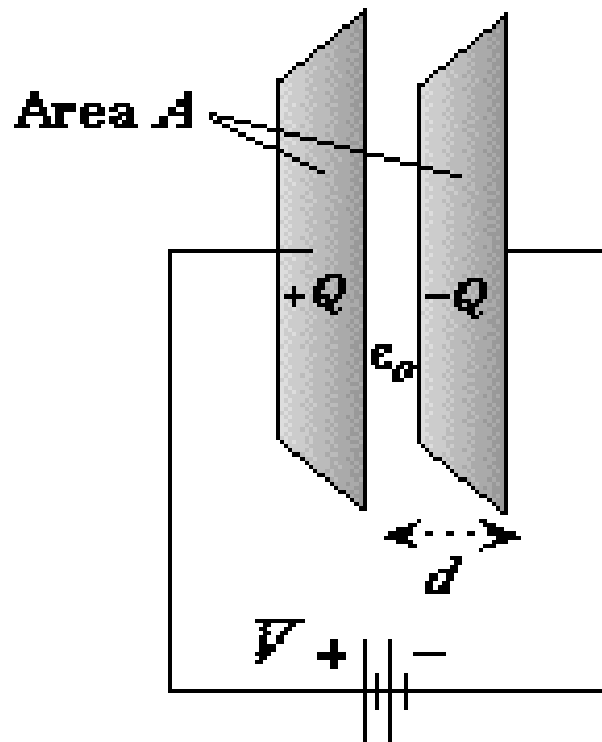
If d changes $\rightarrow$ charges flow $\rightarrow$ electric signal

When the capacitor is connected to the terminals of a battery, electrons transfer between the plates and the wires so that the plates become charged.

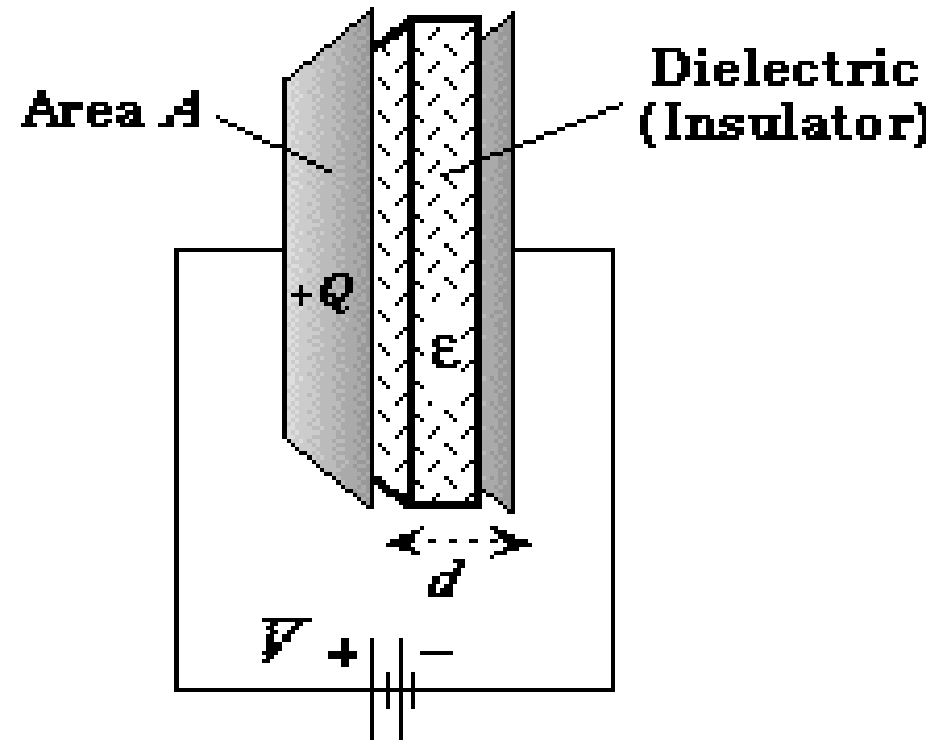


x

$$C = \epsilon_0 \frac{A}{d}$$



$$C = \epsilon \frac{A}{d}$$



Textbook 24.1 and 24.2 do it class

- Which of the following will result in an increase in the capacitance of a parallel-plate capacitor?
 - A. Decreasing the plate area and increasing plate separation
 - B. Increasing the plate area and decreasing the plate separation
 - C. Decreasing the potential difference between the plates
 - D. Increasing the potential difference between the plates

<https://www.youtube.com/watch?v=EoWMF3Vkl6U>
4:43 and 5:23 fun with ultracapacitor



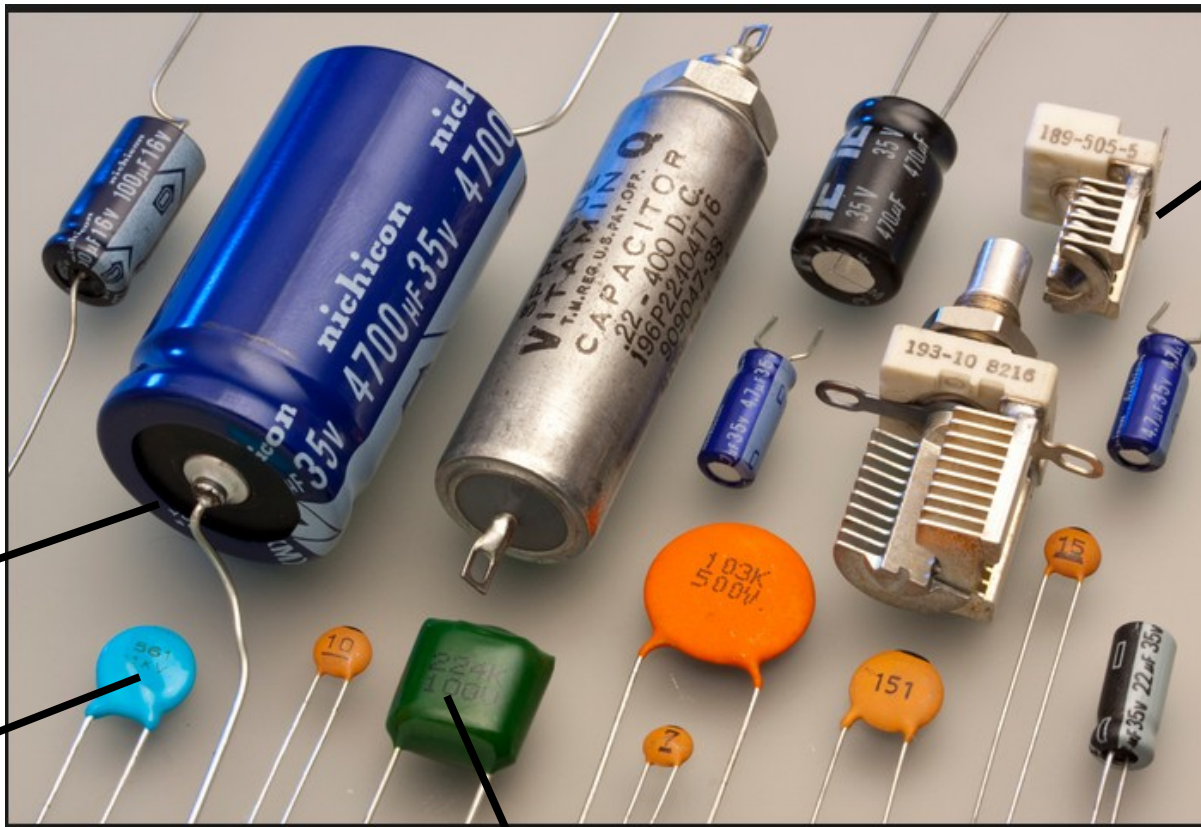
Unroll an electrolytic capacitor



Unroll a super capacitor

<https://www.youtube.com/watch?v=ljeF-l5FpOI>

Inside Look Of A Supercapacitor vs Electrolytic Capacitor



variable

Electrolytic
Larger C

Ceramic
.pF to uF

polyethylene
<10uF

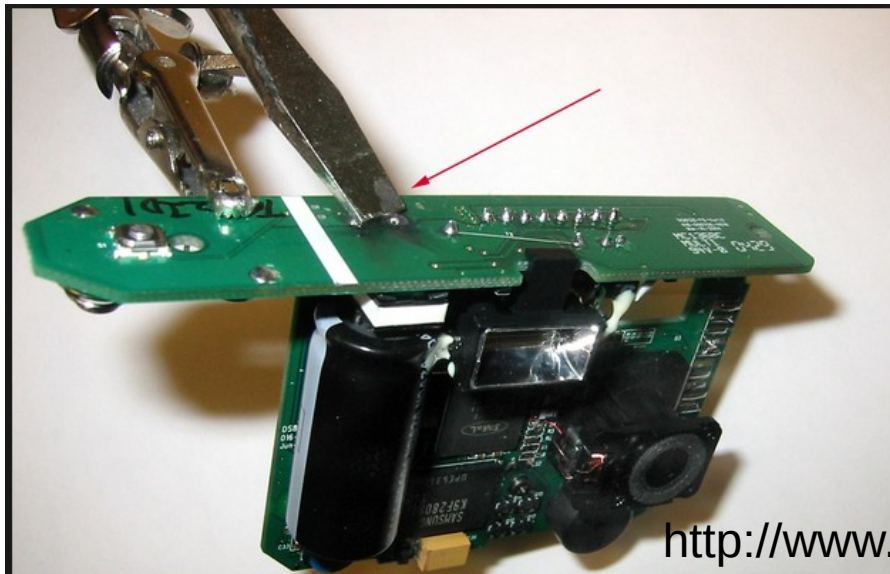


Mica
.pF to uF

<https://electronicspani.com/types-of-capacitors/>



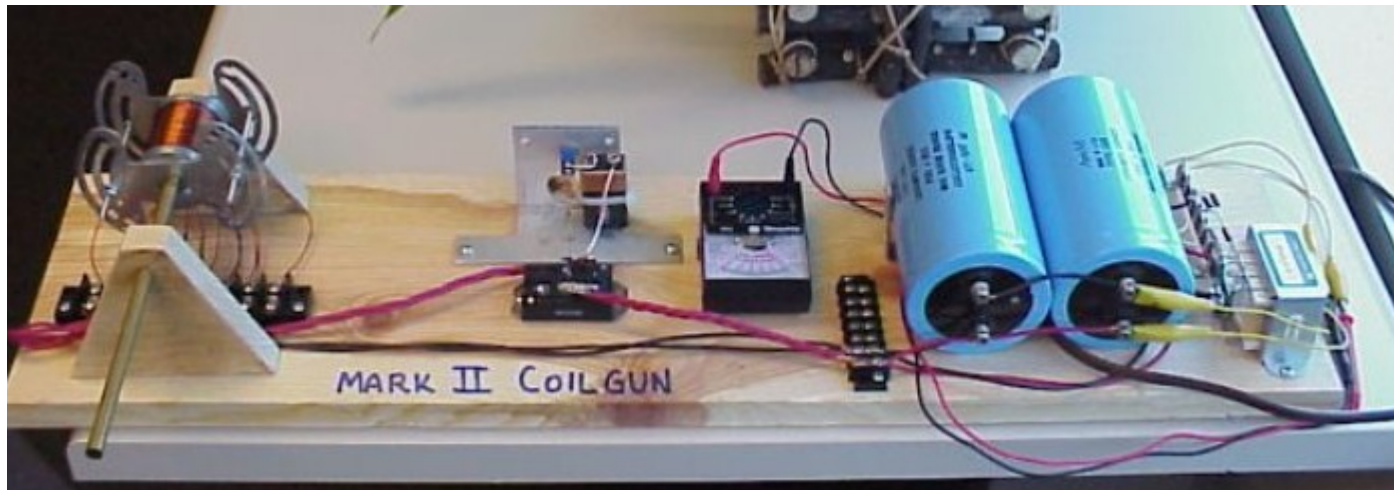
Capacitor bank to power hand made coil gun

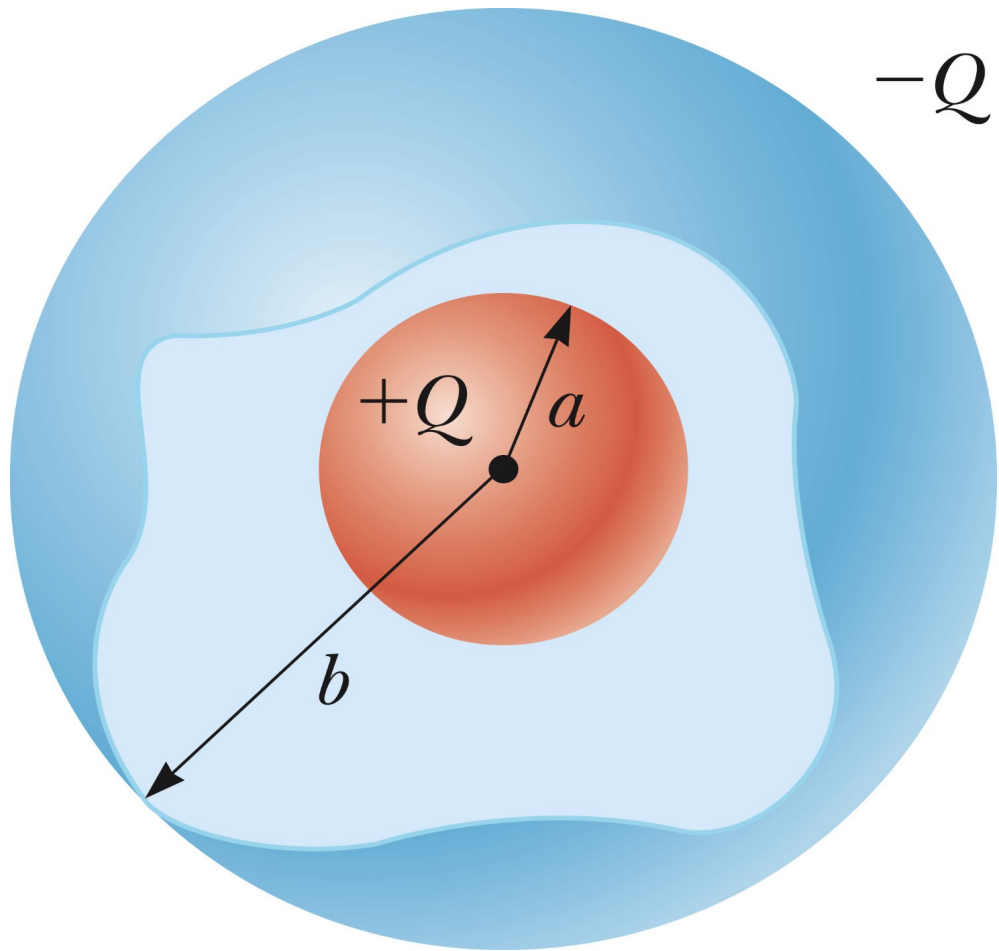


A capacitor can be used to make a coil gun
Some collect it in a disposable
Camera.

3300uF

<http://www.ladyada.net/make/sudc4kap/make.html>



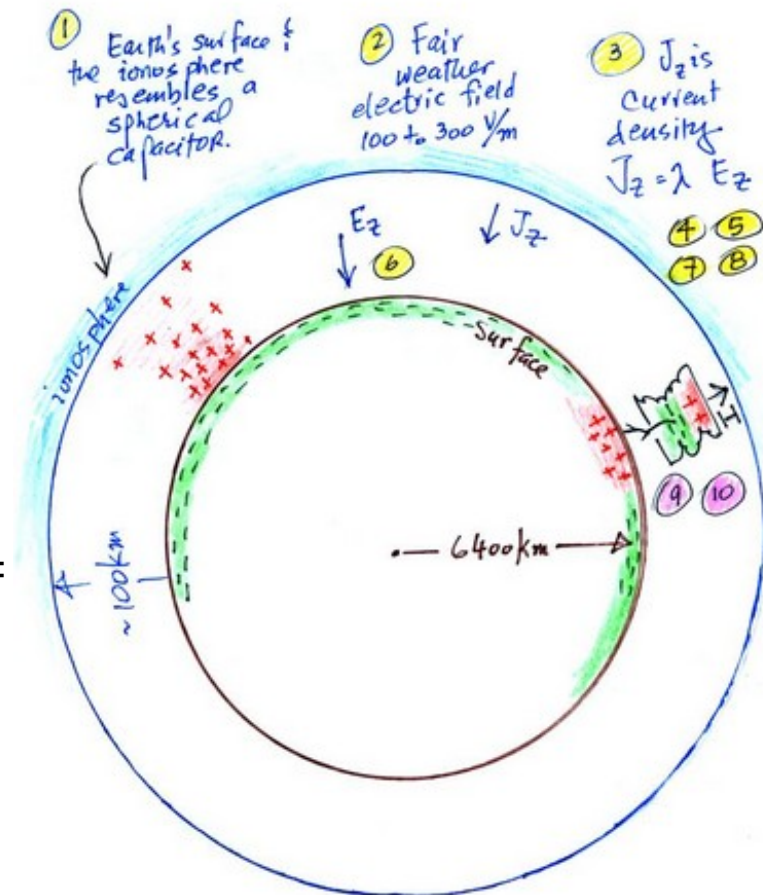


Spherical capacitors.

To find V integrate E between a and b .

$$C = \frac{Q}{\Delta V} = \frac{4\pi\epsilon_0}{\left[\frac{1}{a} - \frac{1}{b}\right]}$$

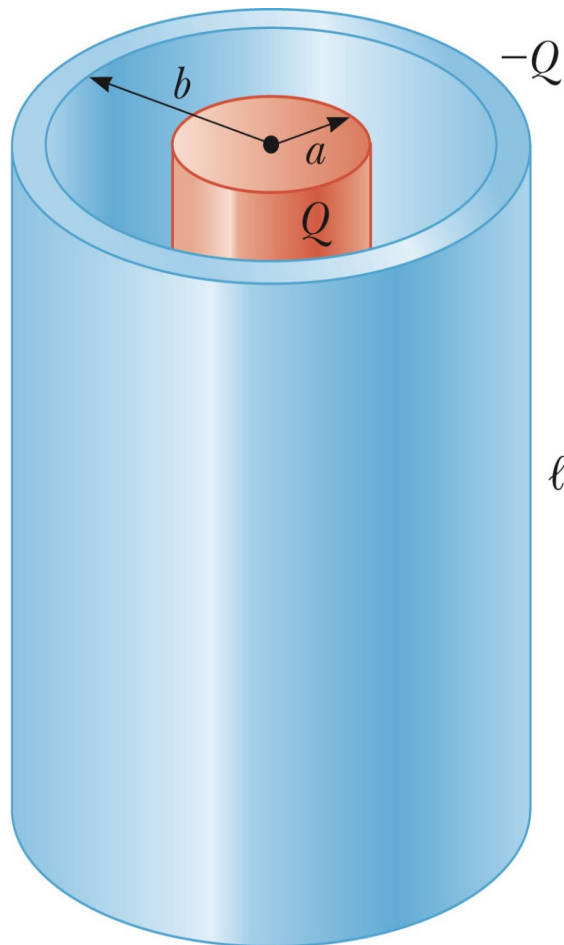
Earth's surface+ionosphere=
Spherical capacitor



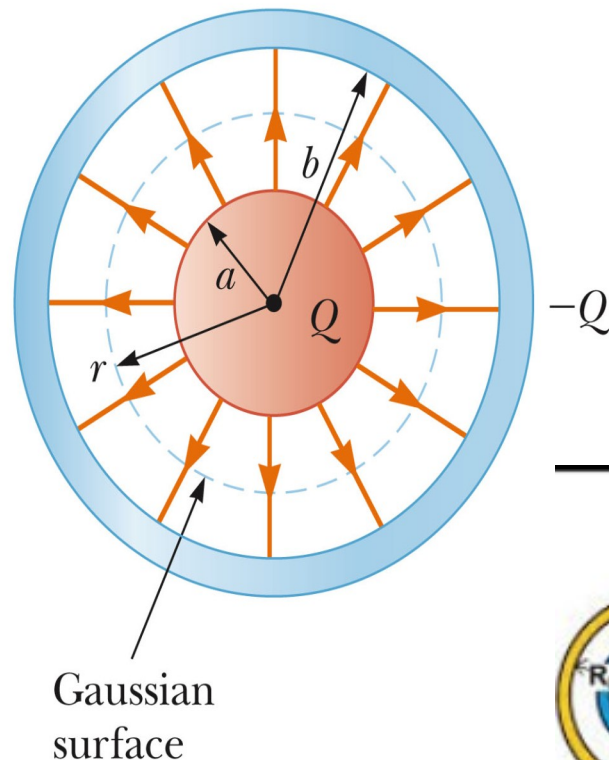
Show how to find C . Then Do textbook 24.3

Cylindrical capacitor

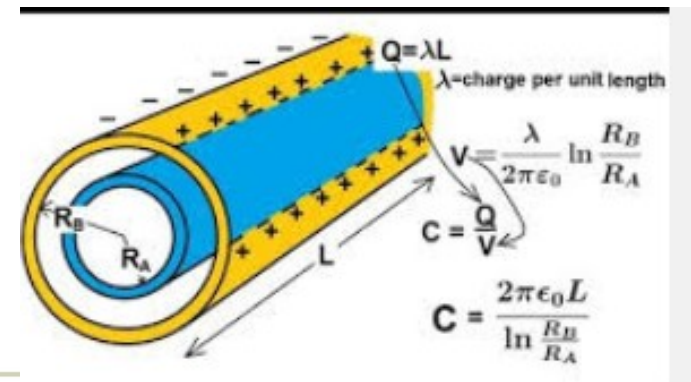
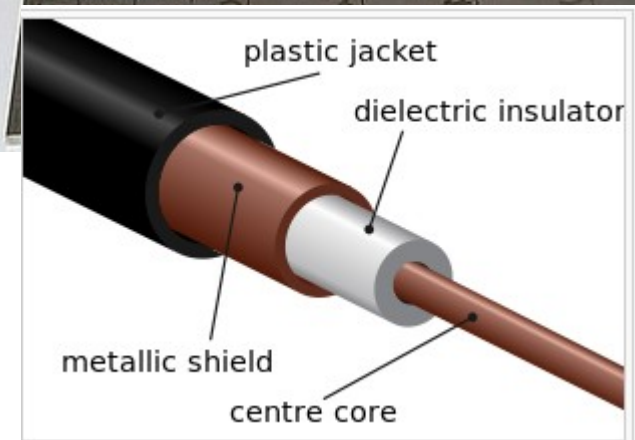
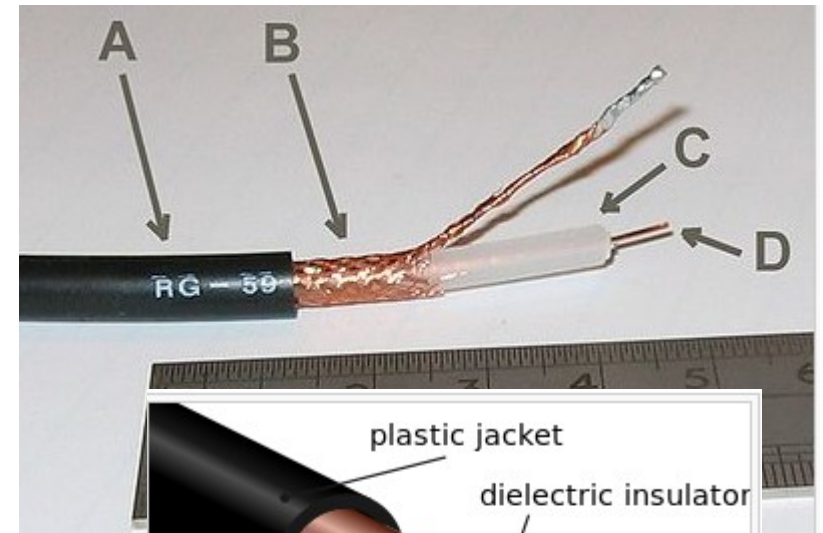
We still use them. If the length is changed, C is changed. See notebook and website above. The coaxial cable used for video signal has a capacitance. The inside is shield from the outside.



a



b



-
- Parallel Plate Capacitor

$$C = \frac{\epsilon_0 A}{d}$$

- Cylindrical (nested cylinder) Capacitor

$$C = 2\pi\epsilon_0 \frac{L}{\ln(b/a)}$$

- Spherical (nested sphere) Capacitor

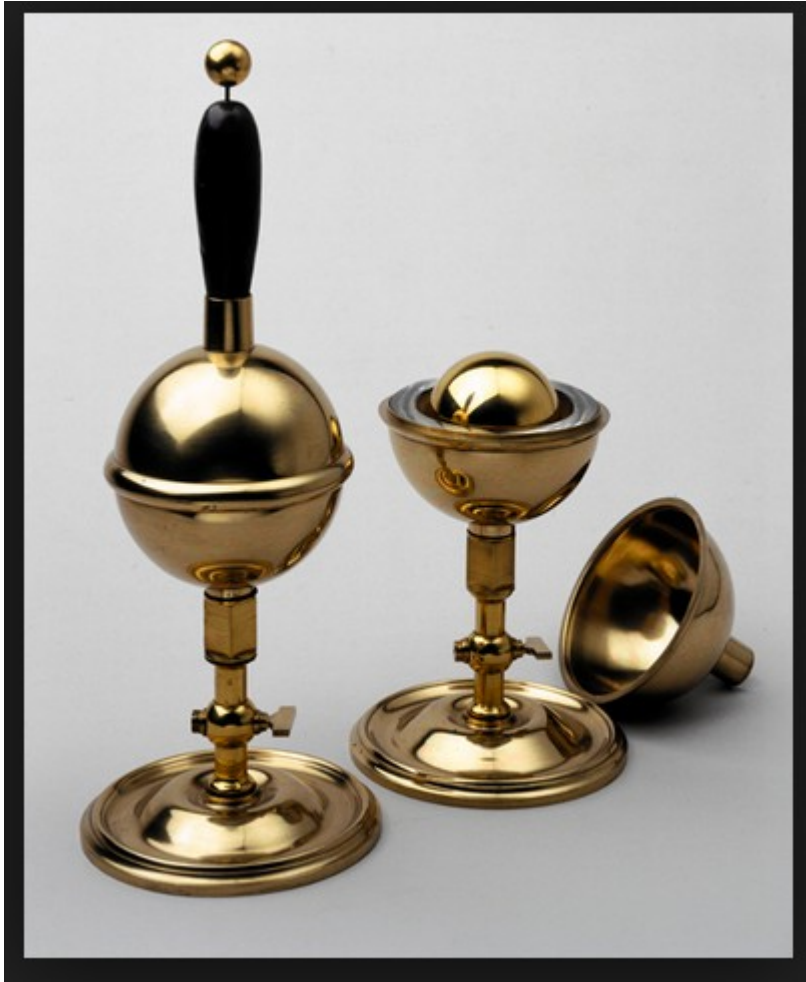
$$C = 4\pi\epsilon_0 \frac{ab}{b-a}$$

- Capacitance for isolated Sphere

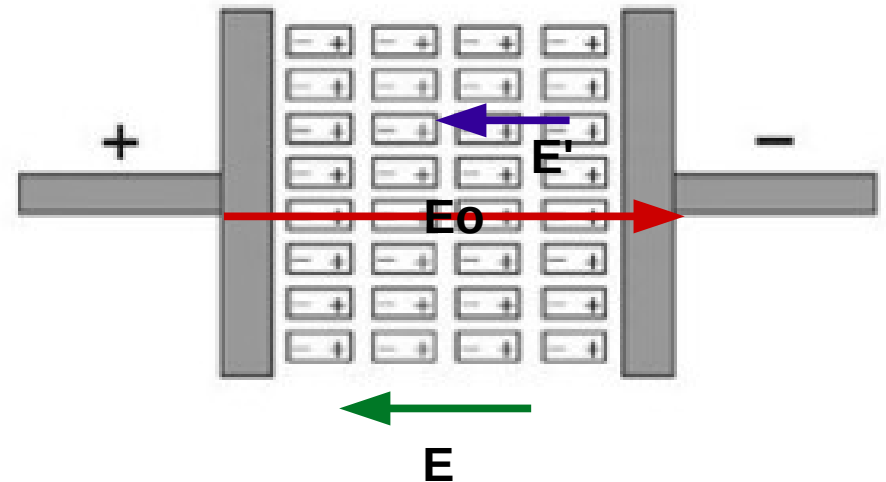
$$C = 4\pi\epsilon_0 R$$

- Units: $\epsilon_0 \times \text{length} = \text{C}^2/\text{Nm} = \text{F}$ (farad), named after Michael Faraday. [note: $\epsilon_0 = 8.85 \text{ pF/m}$]

dielectrics



1791 – 1867



Faraday worked on spherical capacitor with dielectrics between the Conductors. He found that the dielectric (insulator that can be easily polarized) Boosts the capacitance by a constant k that depends on the material.

19.5 Capacitors and Dielectrics

A parallel plate capacitor consists of two metal plates, one carrying charge $+q$ and the other carrying charge $-q$.

It is common to fill the region between the plates with an electrically insulating substance called a **dielectric**.

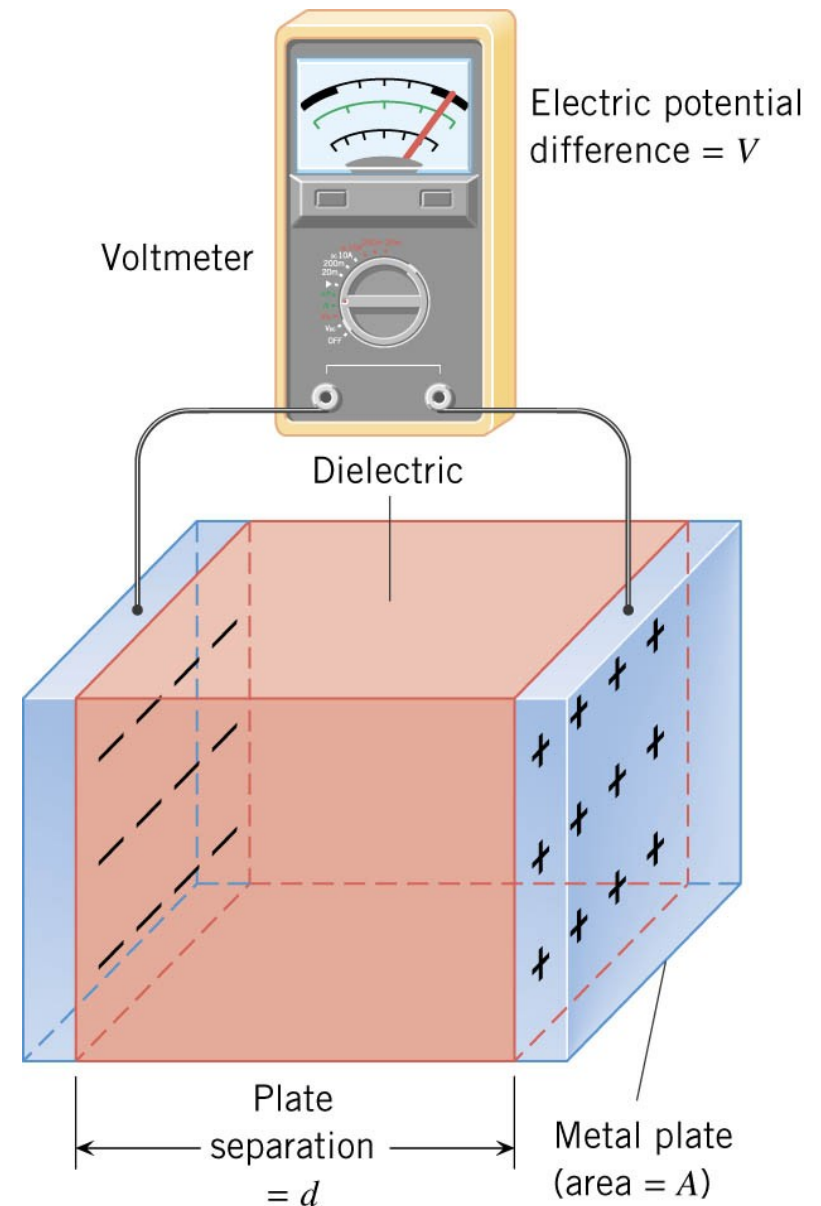
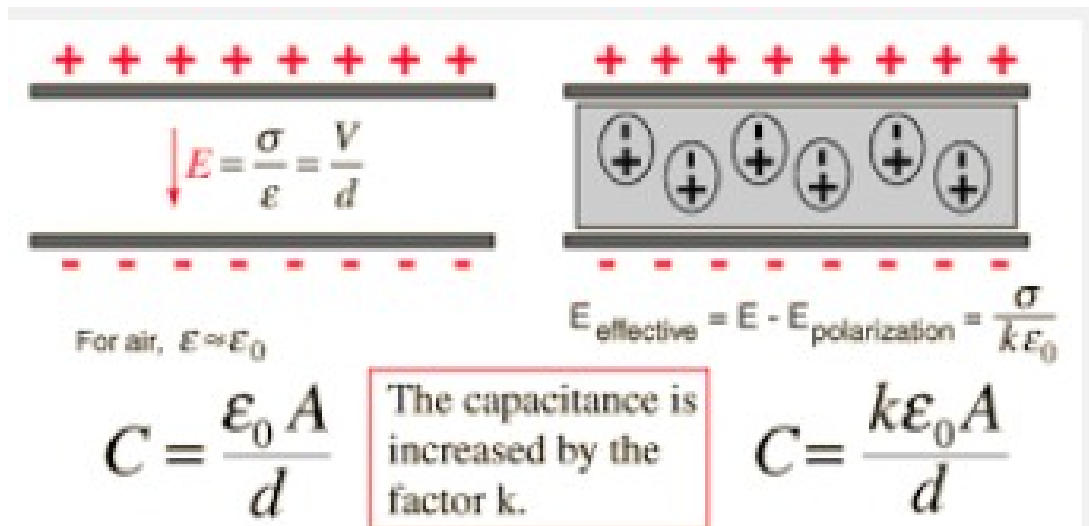


TABLE 17-3 Dielectric constants (at 20°C)

Material	Dielectric constant K	Dielectric strength (V/m)
Vacuum	1.0000	
Air (1 atm)	1.0006	3×10^6
Paraffin	2.2	10×10^6
Polystyrene	2.6	24×10^6
Vinyl (plastic)	2-4	50×10^6
Paper	3.7	15×10^6
Quartz	4.3	8×10^6
Oil	4	12×10^6
Glass, Pyrex	5	14×10^6
Rubber, neoprene	6.7	12×10^6
Porcelain	6-8	5×10^6
Mica	7	150×10^6
Water (liquid)	80	
Strontium titanate	300	8×10^6



K Between 7 and 8

Dielectric Material	Dielectric Constant	Breakdown Field (MV/m)
Air	1.0006	3
Aluminum oxide	8.4	670
Glass (Pyrex)	5.6	14
Paper	3.5	14
Plexiglas	3.4	40
Polyethylene	2.3	50
Polystyrene	2.6	25
Quartz	3.8	8
Tantalum oxide	26	500
Teflon	2.1	60
Water	80	depends on time and purity

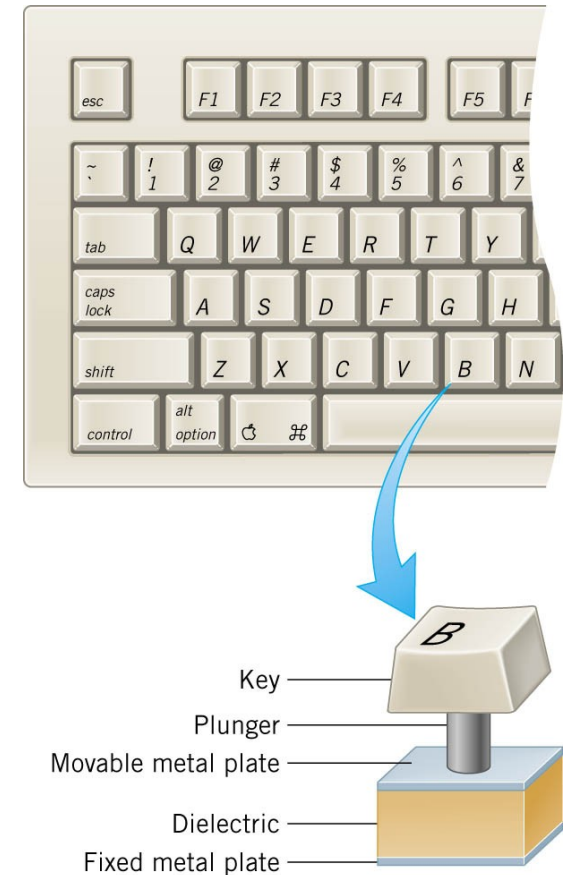
19.5 Capacitors and Dielectrics

Example 12 A Computer Keyboard

One common kind of computer keyboard is based on the idea of capacitance. Each key is mounted on one end of a plunger, the other end being attached to a movable metal plate. The movable plate and the fixed plate form a capacitor. When the key is pressed, the capacitance increases. The *change* in capacitance is detected, thereby recognizing the key which has been pressed.

The separation between the plates is 5.00 mm, but is reduced to 0.150 mm when a key is pressed. The plate area is $9.50 \times 10^{-5} \text{ m}^2$ and the capacitor is filled with a material whose dielectric constant is 3.50.

Determine the change in capacitance detected by the computer.



19.5 Capacitors and Dielectrics

$$C = \frac{\kappa \epsilon_o A}{d} = \frac{(3.50)(8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2))(9.50 \times 10^{-5} \text{ m}^2)}{0.150 \times 10^{-3} \text{ m}} = 19.6 \times 10^{-12} \text{ F}$$

$$C = \frac{\kappa \epsilon_o A}{d} = \frac{(3.50)(8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2))(9.50 \times 10^{-5} \text{ m}^2)}{5.00 \times 10^{-3} \text{ m}} = 0.589 \times 10^{-12} \text{ F}$$

$$\Delta C = 19.0 \times 10^{-12} \text{ F}$$

Always true :

$$C = k C_0 \text{ and } C = Q_{\text{free}}/V$$

C_0 is the capacitance without dielectric

$$E = V/d \text{ and } E = E_0/k \text{ or } E = Q_{\text{free}} / k \epsilon_0$$

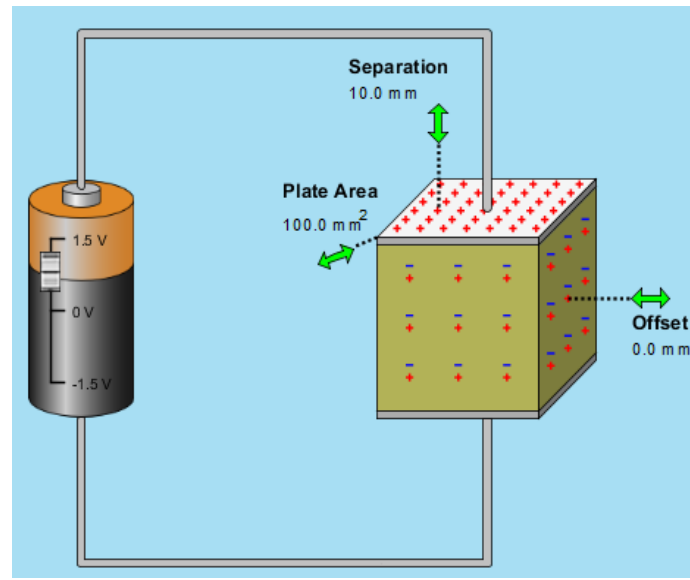
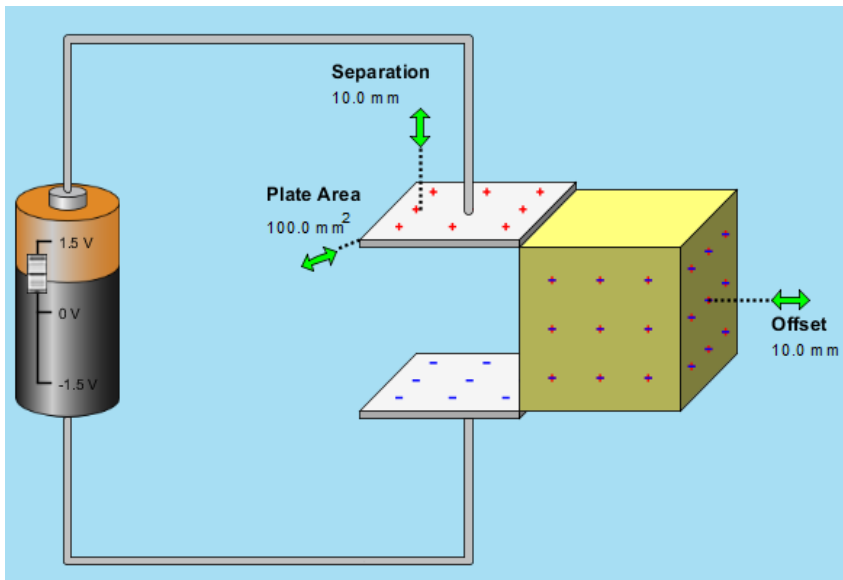
where E is the electric field between the plates and d the distance between them.

$$C = k \epsilon_0 A / d \text{ or } C_0 = \epsilon_0 A / d$$

2 ways :

- 1) the capacitor stays connected to the battery and the dielectric is inserted.
In that case the voltage has to stay constant. But the charge increases

<https://phet.colorado.edu/en/simulation/capacitor-lab>



$C = k C_0$ the capacitance increases by a factor of k

$V = V_0$ stays the same so E stays the same. (Q_{free} increases so the electric field increases

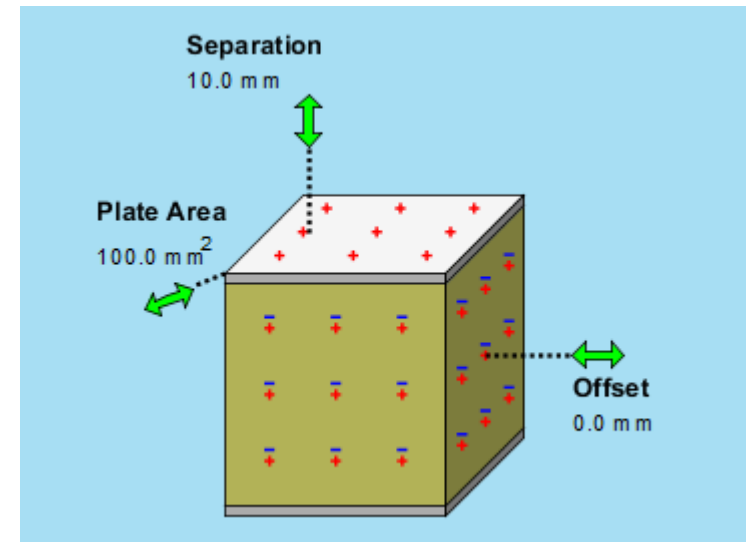
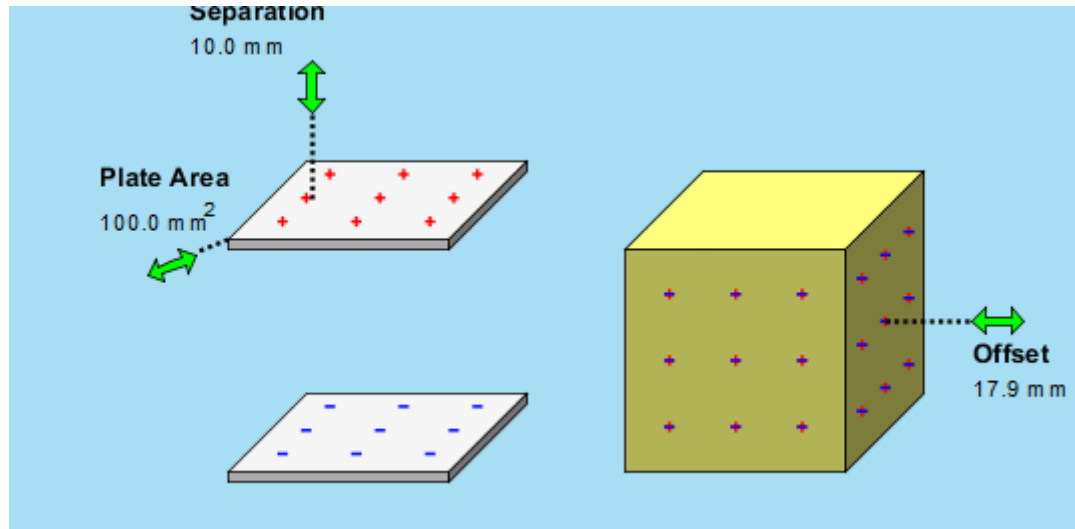
But at the same time the electric field in the dielectric fights it)

$Q_{\text{free}} = V_0 C = k V_0 C_0$ so Q_{free} increases by a factor k

The energy is $\frac{1}{2} C V^2$. So the energy increases by a factor k . This factor comes from

You pushing the dielectric in !

2) The capacitor is charged and disconnected. Then you place the dielectric. The charges have nowhere to go. So $Q = Q_0$. The voltage can change as the electric field can change.



The charge is the same $Q = Q_0$ and $C = kC_0$

$V = Q/C = Q_0/C = Q_0/kC_0$ so $V = V_0/k$. The voltage decreases by a factor k .

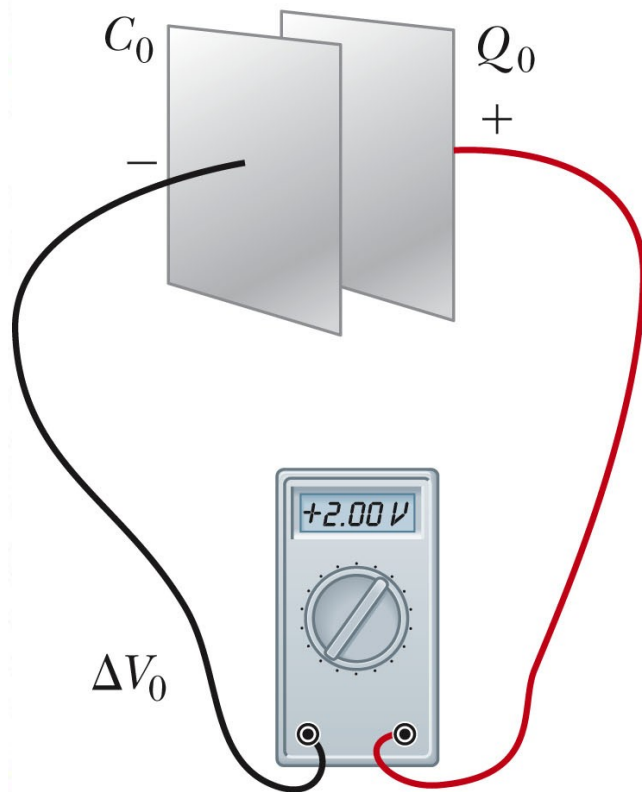
$E = V/d = V_0/kd$ so the electric field decreases by a factor k because of the dielectric's own electric field and the charge stays constant.

$U = \frac{1}{2} C V^2 = \frac{1}{k} U_0$ So the energy decreases !! That means the dielectric is sucked in !

6. dielectric

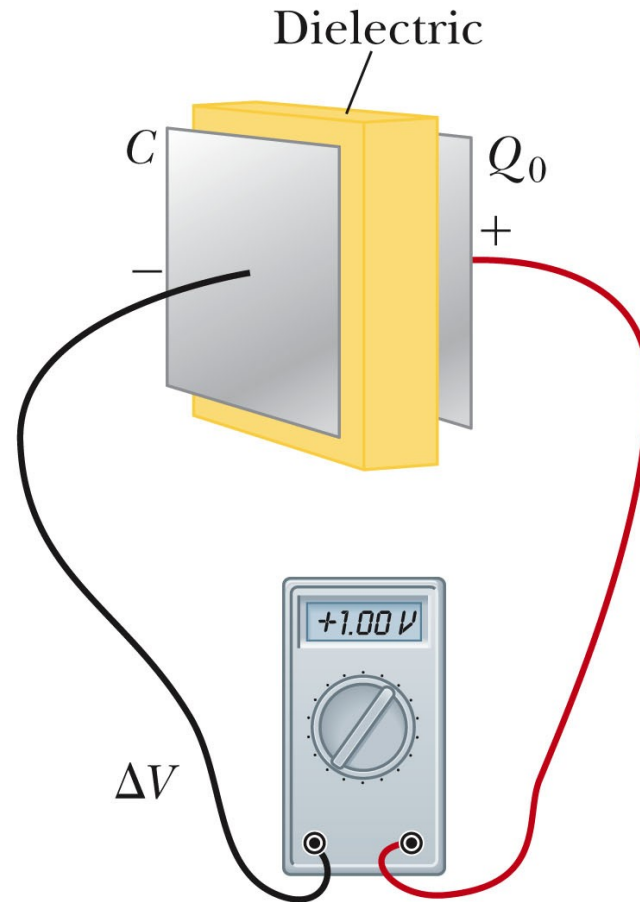
<http://phet.colorado.edu/en/simulation/capacitor-lab>

The potential difference across the charged capacitor is initially ΔV_0 .



a

After the dielectric is inserted between the plates, the charge remains the same, but the potential difference decreases and the capacitance increases.

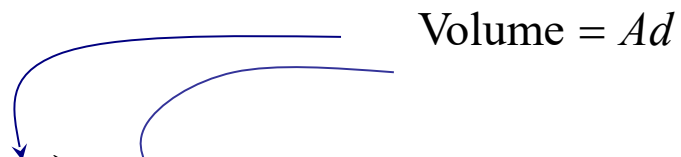


b

Fig. 26-13, p. 753

ENERGY STORAGE IN A CAPACITOR

$$\text{Energy} = \frac{1}{2} CV^2$$



Volume = Ad

$$\text{Energy} = \frac{1}{2} \left(\frac{\kappa \epsilon_o A}{d} \right) (Ed)^2$$

$$\text{Energy density} = \frac{\text{Energy}}{\text{Volume}} = \frac{1}{2} \kappa \epsilon_o E^2$$

So we can find the energy stored in any volume containing an electric field

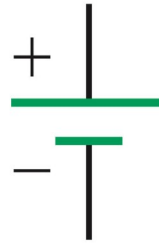
Textbook 24.8 and 24.9 Use slide 22 to find the energy stored In a spherical capacitor.

- You need to replace a capacitor with one that can store more electrical energy. Which one of the following will give you a greater energy increase?
 - A. A capacitor with twice the capacitance and the same working voltage.
 - B. A capacitor with twice the working voltage and the same capacitance.
 - C. A capacitor with half the capacitance and twice the working voltage.

Capacitor
symbol



Battery
symbol



Switch
symbol

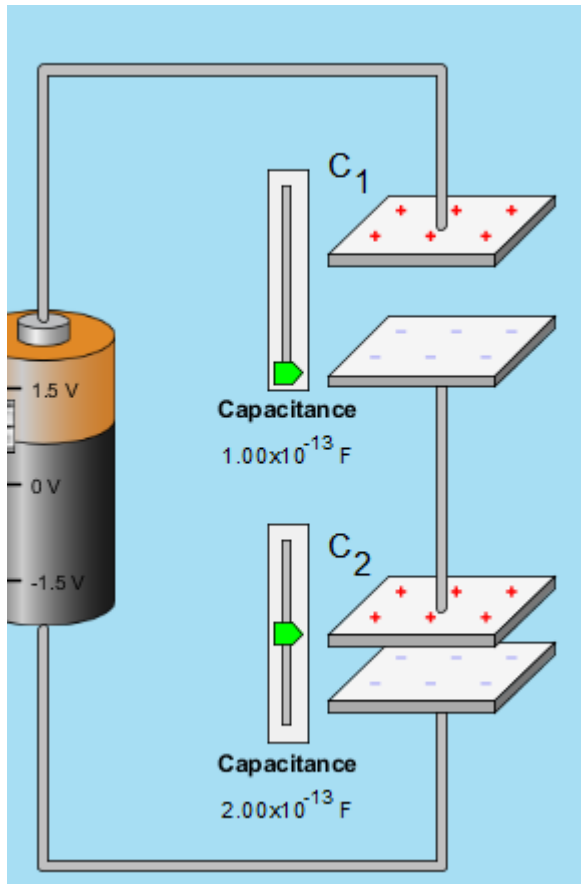


Open

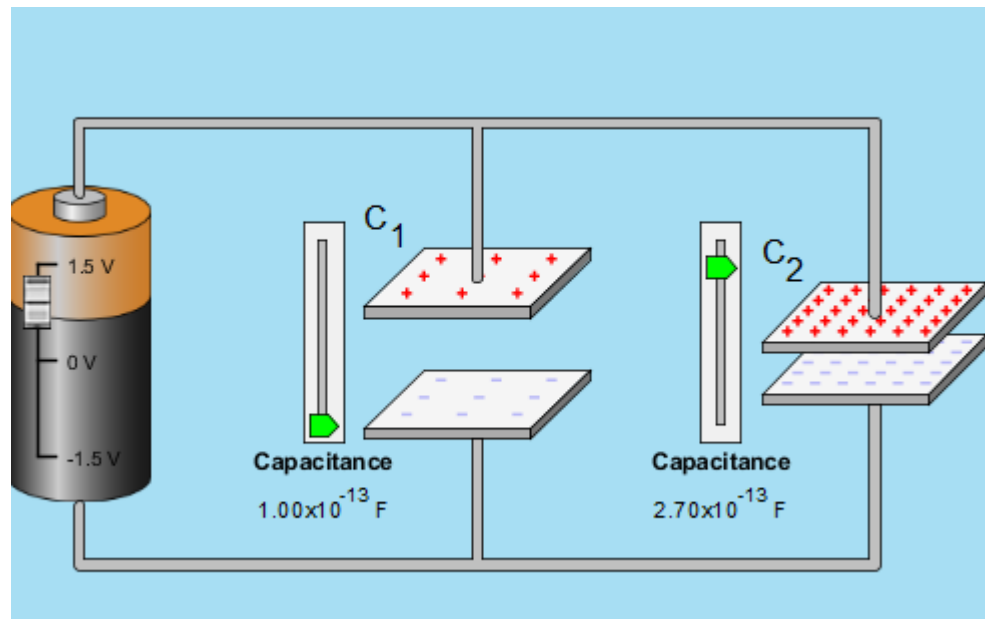


Closed

Notice how the capacitance is not
The same but the charge is the same.
The voltage is not the same.
 $V=Q/C$ its inversely proportional to C



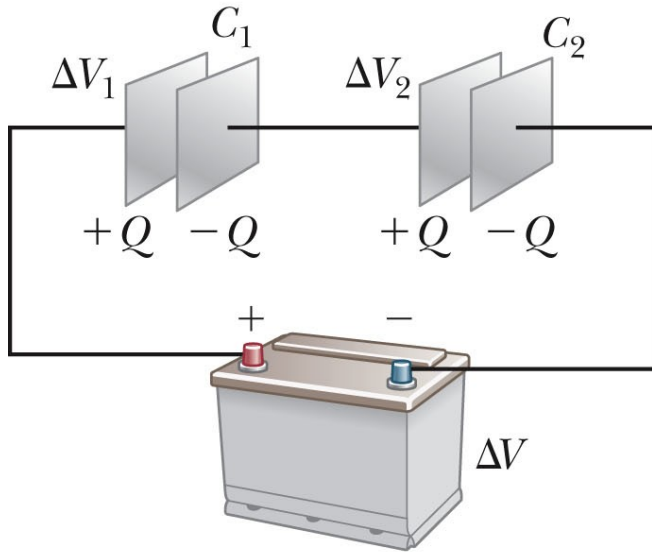
The capacitors are in parallel.
Notice that the charge is not the same.
The capacitance is not the same.
But the voltage is the same !



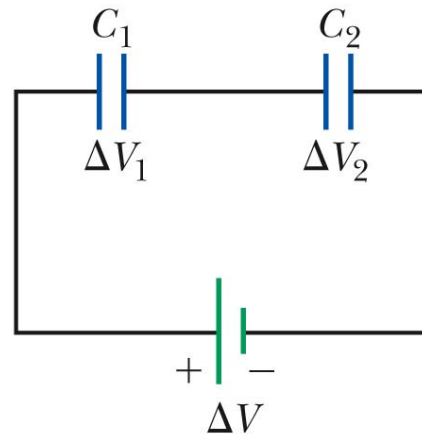
<https://phet.colorado.edu/en/simulation/capacitor-lab>

Capacitor in series

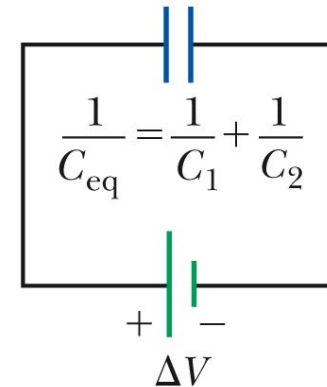
A pictorial representation of two capacitors connected in series to a battery



A circuit diagram showing the two capacitors connected in series to a battery



A circuit diagram showing the equivalent capacitance of the capacitors in series

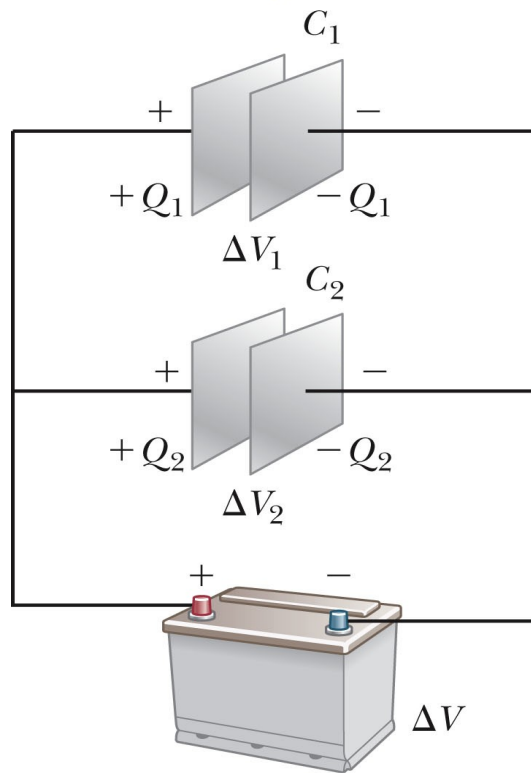


a Equivalent capacitance = capacitance of a single one that can be substituted for the Combination of capacitors with no change in the operation of the external circuit.

<https://phet.colorado.edu/en/simulation/capacitor-lab>

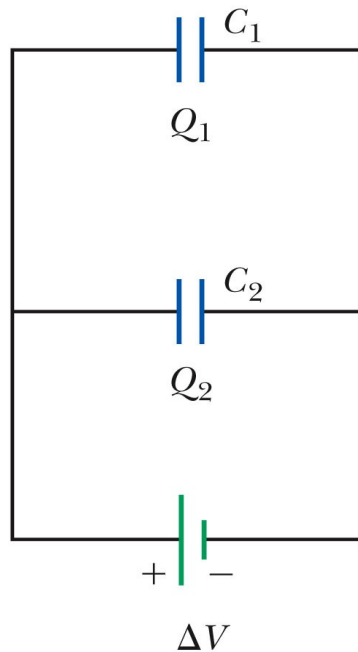
Capacitor in parallel

A pictorial representation of two capacitors connected in parallel to a battery



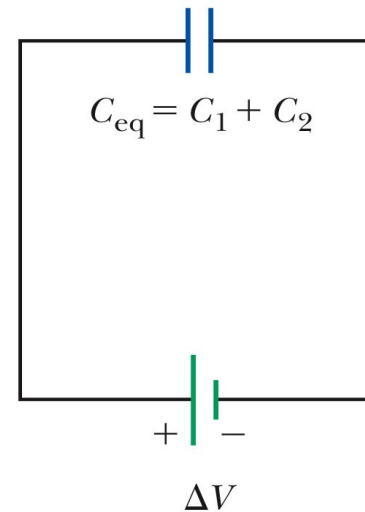
a

A circuit diagram showing the two capacitors connected in parallel to a battery



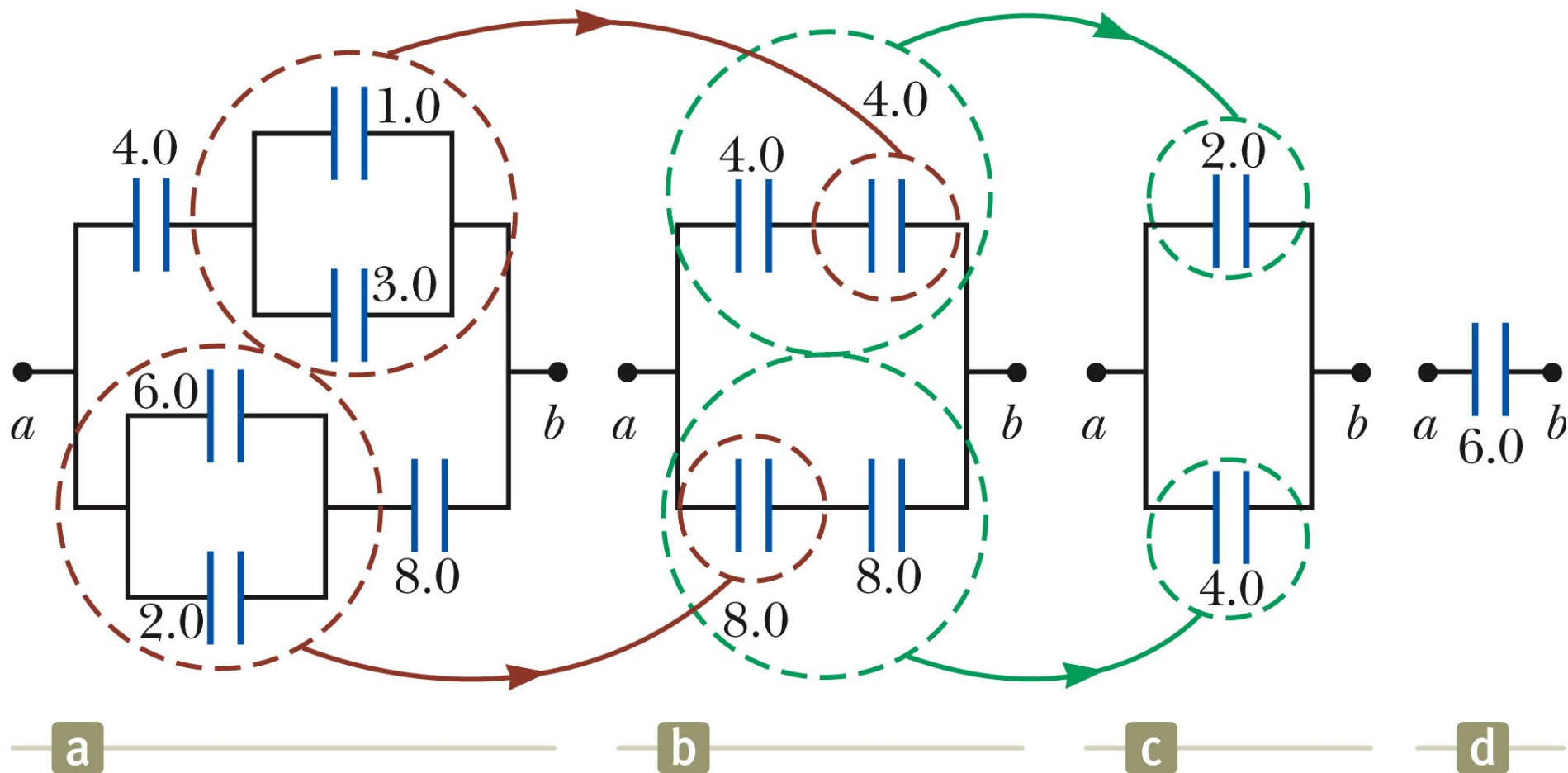
b

A circuit diagram showing the equivalent capacitance of the capacitors in parallel

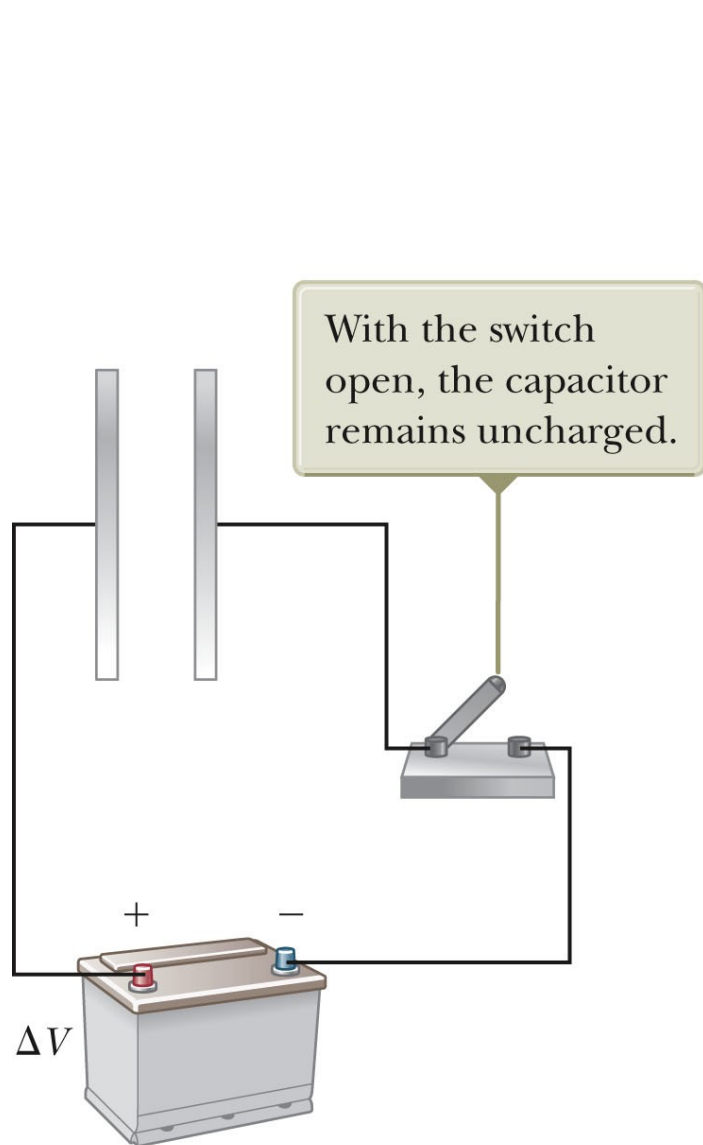


c

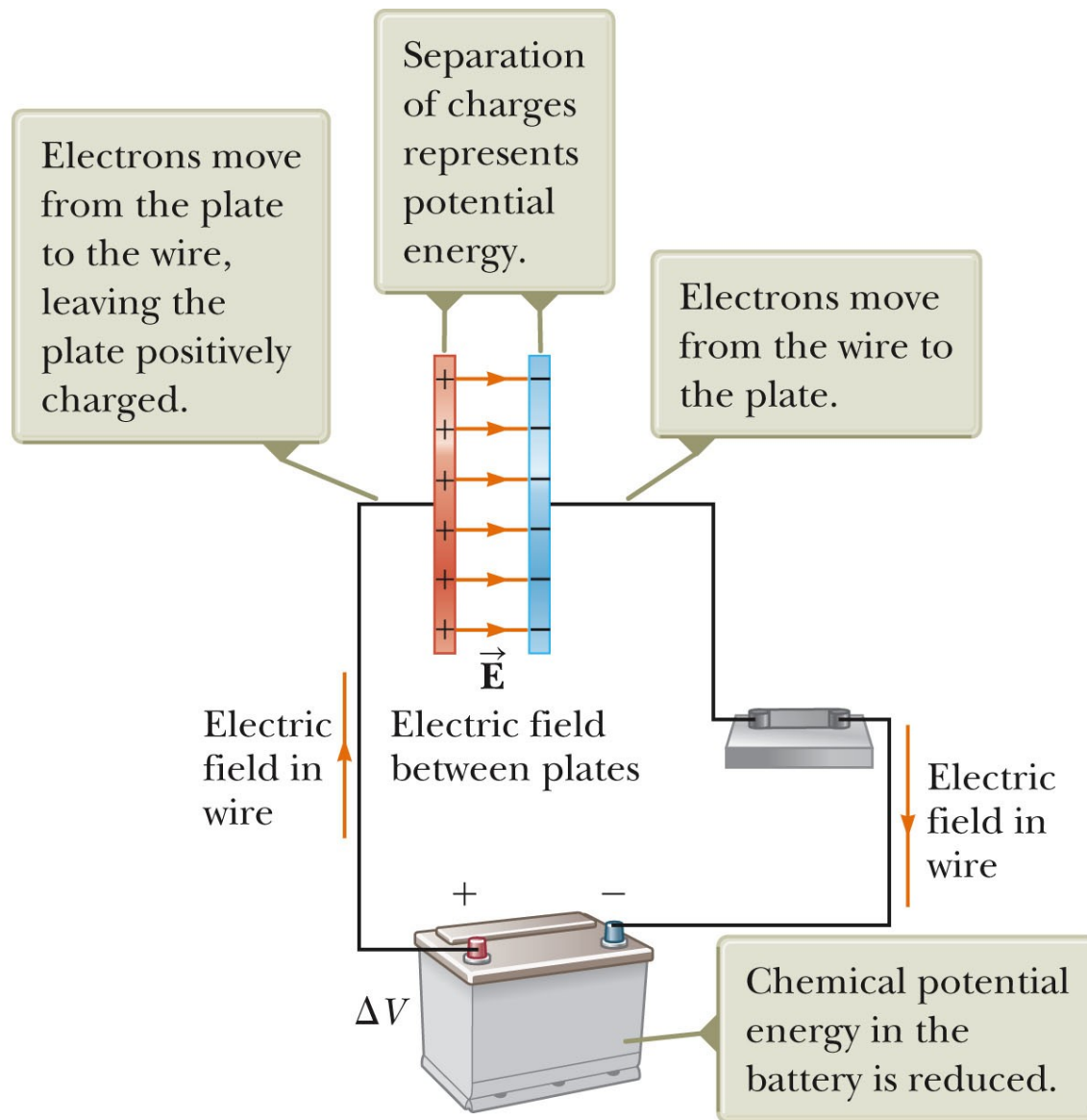
Do textbook 24.5



Textbook 24.6 do and 24.7



a



b